

Differential Geometry Notes

Aditya Padiyar

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1 The derivative

Claim 1. *The map $X, Y \rightarrow D_X Y$ is bilinear and has the following properties:*

- $D_{fX} Y = f(D_X Y)$
- $D_X(fY) = f(D_X Y) + (D_X f)Y$
- $D_X Y - D_Y X = [X, Y]$
- $D_X \langle Y, Z \rangle = \langle D_X Y, Z \rangle + \langle Y, D_X Z \rangle$

Claim 2. *Suppose $\phi : \Omega \rightarrow V$ is a local parametrization. Then $D_{\phi_i} \phi_j = \frac{\partial \phi}{\partial x_i \partial x_j}$*

Suppose $F : M \rightarrow N$ is a diffeomorphism

Claim 3. *Suppose $v \in T_p M$. Then for all $f \in C^\infty(M)$ we have $D_v(f \circ F) = (D_{F_* v})f$*

Claim 4. *We have $F_*[X, Y] = [F_* X, F_* Y]$*

Claim 5. *We have $\langle F_* X, F_* Y \rangle = \langle X, Y \rangle$*

2 Riemannian metric

Definition 1. *Suppose $\gamma : [a, b] \rightarrow M$ is a piecewise smooth curve.*

A reparametrization of γ is given by $\gamma \circ \phi$, where $\phi : [c, d] \rightarrow [a, b]$ is a homeomorphism with a partition $c_0, c_1, \dots, c_k$ such that $\phi|_{[c_{i-1}, c_i]}$ is a diffeomorphism onto its image

Claim 6. *Suppose M is connected. Then there is a piecewise regular curve between any two points*

Definition 2. *Suppose M is connected. Then we define the Riemannian metric by*

$$d_M(p, q) = \inf\{L(\gamma) : \gamma \text{ is a piecewise smooth curve from } p \text{ to } q\}$$

Claim 7. *The topology induced by the Riemannian metric is the same as the subspace topology M inherits from $\mathbb{R}^k$*

Claim 8. If there is a piecewise smooth curve from p to q of length $d_M(p, q)$ then there is such a piecewise regular curve as well. This is true for every p and q if M is closed in $\mathbb{R}^k$

Definition 3. A diffeomorphism $F : M \rightarrow N$ is called a Riemannian isometry if at each point the derivative DF_p is an isometry from T_pM to T_pN

Claim 9. Every Riemannian isometry is an isometry with respect to the Riemannian metric

Definition 4. A smooth map $F : M \rightarrow N$ is called a local Riemannian isometry at p if there is a neighborhood U such that $F(U)$ is open and $F|_U : U \rightarrow F(U)$ is a Riemannian isometry

3 First fundamental form

Definition 5. The first fundamental form of M at p is defined by restricting the standard inner product on $\mathbb{R}^k$ to the tangent space. We have $g_p : T_pM \times T_pM \rightarrow \mathbb{R}$

Claim 10. Two manifolds M and N admit a local Riemannian isometry from p to q if and only if there are local parametrizations $\phi : \Omega \rightarrow U$ and $\varphi : \Omega \rightarrow V$ such that $g_{\phi(a)}(v_i, v_j) = h_{\varphi(a)}(w_i, w_j)$ for all $a \in \Omega$, where $v_i = \phi_i(\phi(a))$ and $w_i = \varphi_i(\varphi(a))$

4 Covariant derivative

Definition 6. We have $\pi_p \in \text{End}(\mathbb{R}^k)$ which is the orthogonal projection onto T_pM . We define $\hat{\pi}_p$ as $I - \pi_p$

Definition 7. Suppose $v \in T_pM$. Define $\nabla_v \in L(\mathcal{X}(M), T_pM)$ by $X \rightarrow \pi_p(DX)_p v$

Definition 8. Suppose $X, Y \in \mathcal{X}(M)$ are smooth vector fields. Then we have the smooth vector field $\nabla_X Y = (p \rightarrow \pi_p(D_X Y)(p))$

Claim 11. The map $X, Y \rightarrow \nabla_X Y$ is bilinear and has the following properties:

- $\nabla_X Y - \nabla_Y X = [X, Y]$
- $\nabla_{fX} Y = f(\nabla_X Y)$
- $\nabla_X(fY) = f(\nabla_X Y) + (D_X f)Y$

Definition 9. Suppose $\phi : \Omega \rightarrow V$ is a local parametrization. The Christoffel symbols $\Gamma_{ij}^k \in C^\infty(\Omega)$ are defined by $(\nabla_{\phi_i} \phi_j)(\phi(a)) = \sum_k \Gamma_{ij}^k(a) \phi_k(a)$

Definition 10. The curvature tensor of X, Y, Z is defined by $R(X, Y, Z) = \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X, Y]} Z$

Suppose $F : M \rightarrow N$ is a Riemannian isometry

Claim 12. *We have $\nabla_{F_*X}(F_*Y) = F_*(\nabla_X Y)$*

Claim 13. *We have $\nabla_{F_*v} \circ F_* = DF_p \circ \nabla_v$*

Claim 14. *We have $F_*[R(X, Y, Z)] = R(F_*X, F_*Y, F_*Z)$*

5 Vector field along a curve

Definition 11. *Suppose $\gamma : I \rightarrow M$ is a smooth curve. A smooth vector field along γ is a map $V : I \rightarrow \mathbb{R}^k$ such that $V(t) \in T_{\gamma(t)}M$*

The vector space of smooth vector fields along γ is $\mathcal{X}(\gamma)$

Definition 12. *We have $\text{cov}_\gamma \in L(\mathcal{X}(\gamma))$ given by $V \rightarrow (t \rightarrow \pi_{\gamma(t)} \dot{V}(t))$*

Definition 13. *A vector field V along γ is called parallel if $V \in \ker(\text{cov}_\gamma)$*

Claim 15. *Suppose $t_0 \in I$, $p_0 = \gamma(t_0)$ and $v_0 \in T_{p_0}M$. There is a unique parallel vector field V satisfying $V(t_0) = v_0$*

6 Second fundamental form

Definition 14. *The second fundamental form of M at p is the bilinear map $S_p : T_pM \times T_pM \rightarrow \mathbb{R}^k$ given by $v, w \rightarrow [(D\pi)_p v]w$*

Claim 16. *range $S_p \subseteq N_pM$ for all p*

Claim 17. *The second fundamental form is symmetric*

Claim 18. *We have $S_p(X_p, Y_p) = \hat{\pi}_p((D_X Y)(p))$*